Meteor Forward-Scatter Doppler: Key Equations

WDPN-TV ch. 2 pilot (54.309441 MHz) → receiver at 38.8217°N, 77.1037°W · from `meteor_doppler_wdpn.py`

1. Setup

$$\lambda = c / f = 299\,792\,458 / 54.309441\times10^{6} = 5.520\ \text{m}$$ wavelength

$$R = |\mathbf{P}-\mathbf{T}| + |\mathbf{P}-\mathbf{R}|$$ bistatic path

T = transmitter, **R** = receiver, **P** = meteor reflection point (all positions in Earth-centred ECEF).

$$\mathbf{u}_\mathrm{T} = (\mathbf{P}-\mathbf{T}) / |\mathbf{P}-\mathbf{T}| \quad \mathbf{u}_\mathrm{R} = (\mathbf{P}-\mathbf{R}) / |\mathbf{P}-\mathbf{R}|$$ unit vectors, station → P

$$\mathbf{b} = \mathbf{u}_\mathrm{T} + \mathbf{u}_\mathrm{R}$$ bisector vector

2. Doppler shift

The echo frequency shift is the rate of change of the path length, in wavelengths per second. With the meteor moving at velocity $\mathbf{v} = v\,\hat{\mathbf{v}}$, $d\mathbf{P}/dt = \mathbf{v}$ and $d|\mathbf{P}-\mathbf{T}|/dt = \mathbf{u}_\mathrm{T}\cdot\mathbf{v}$, so:

$$f_\mathrm{D} = -\frac{1}{\lambda}\frac{\mathrm{d}R}{\mathrm{d}t} = -\frac{v}{\lambda}\left(\hat{\mathbf{v}}\cdot\mathbf{b}\right)$$ core result

Sign convention: $f_\mathrm{D} > 0$ means the path is shortening (echo shifted up).

3. Meteor velocity direction (local East-North-Up at P)

$$\hat{\mathbf{v}} = \cos d \sin(\mathrm{az})\,\hat{\mathbf{e}} + \cos d \cos(\mathrm{az})\,\hat{\mathbf{n}} - \sin d\,\hat{\mathbf{u}}$$ heading az, descent d

az = direction of travel measured clockwise from north; d = angle below the horizontal. Dotting with **b** gives components b_e, b_n, b_u:

$$f_D(\mathrm{az}, d) = -\frac{v}{\lambda}\,[\,\cos d\,(\,b_e \sin \mathrm{az} + b_n \cos \mathrm{az}\,) - \sin d\, b_u\,]$$

4. Maximum Doppler and the zero-Doppler surface

peak Doppler

$$|f_D|_{\max} = \frac{v}{\lambda}\,|\mathbf{b}|\ ,\quad |\mathbf{b}| = 2\cos(\theta/2)$$

θ = angle between $\mathbf{u}_T$ and $\mathbf{u}_R$. Reached when the meteor moves along **b**.

zero Doppler

$$f_D = 0 \iff \hat{\mathbf{v}} \perp \mathbf{b}$$

Geometrically: the meteor's track is tangent to the ellipsoid of constant path length (foci at T and R) through P.

5. Zero-Doppler descent angle for a given heading

$$h = b_e \sin \mathrm{az} + b_n \cos \mathrm{az} \ \Rightarrow\ \cos d\, h - \sin d\, b_u = 0 \ \Rightarrow\ d_0 = \mathrm{atan2}(\,h, b_u\,)$$

Folded into −90°…+90°; kept only if $0° \le d_0 \le 90°$ (a meteor can't climb).

6. The ±100 Hz detection band

fraction of the unit sphere

$$|\hat{\mathbf{v}} \cdot \hat{\mathbf{b}}| \le k = \frac{f_{\lim}\,\lambda}{v\,|\mathbf{b}|}\ ,\quad f_{\lim} = 100\ \mathrm{Hz}$$

A thin band of half-angle arcsin *k* around the great circle of directions perpendicular to **b**. Its share of random directions is about *k* (a band of angular half-width arcsin *k* covers a fraction *k* of the sphere's area).

7. Headings giving 0 Hz and ±100 Hz at a fixed descent angle

$$b_h = \sqrt{(b_e{}^2 + b_n{}^2)}\ ,\quad \mathrm{az}_b = \mathrm{atan2}(b_e, b_n)\ \Rightarrow\ h = b_h \cos(\mathrm{az} - \mathrm{az}_b)$$

$$\cos(\mathrm{az} - \mathrm{az_b}) = c_0 = \frac{b_u \tan d}{b_h} \;\Rightarrow\; \mathrm{az} = \mathrm{az_b} \pm \arccos c_0$$

zero-Doppler headings

$$c = c_0 \pm \Delta c\,, \quad \Delta c = \frac{f_{\mathrm{lim}}\,\lambda}{v \cos d\, b_h} \;\Rightarrow\; \mathrm{az} = \mathrm{az_b} \pm \arccos c$$

±100 Hz edges

Two mirror-image solutions (the ±) exist whenever $|c| \le 1$; otherwise no heading gives that Doppler at that descent angle.

8. Geometry helpers

$$N = \frac{a}{\sqrt{(1 - e^2 \sin^2\varphi)}}$$

$$\mathbf{x} = \left((N+h)\cos\varphi\cos\lambda_o\,,\ (N+h)\cos\varphi\sin\lambda_o\,,\ (N(1-e^2)+h)\sin\varphi \right)$$

geodetic → ECEF (WGS-84)

a = 6 378 137 m, e^2 = 0.00669438; φ = latitude, λ_o = longitude.

$$\hat{\mathbf{e}} = (-\sin\lambda_o, \cos\lambda_o, 0)$$

$$\hat{\mathbf{n}} = (-\sin\varphi\cos\lambda_o, -\sin\varphi\sin\lambda_o, \cos\varphi)$$

$$\hat{\mathbf{u}} = (\cos\varphi\cos\lambda_o, \cos\varphi\sin\lambda_o, \sin\varphi)$$

local ENU basis

$$\mathbf{g} = (1-s)\mathbf{T}_0 + s\mathbf{R}_0\,, \quad \hat{\mathbf{g}} = \mathbf{g}/|\mathbf{g}|\,, \quad \mathrm{lat} = \mathrm{atan2}\left(g_z\,,\ (1-e^2)\sqrt{(g_x{}^2 + g_y{}^2)} \right)$$

reflection point above the path

s = fraction along the path (0.5 = midpoint); $\mathbf{T}_0$, $\mathbf{R}_0$ are the station positions at zero altitude; the point is then placed at the chosen meteor altitude (80–120 km). Off-baseline points add $x\cdot\hat{\mathbf{t}} + y\cdot(\hat{\mathbf{u}}\times\hat{\mathbf{t}})$, with $\hat{\mathbf{t}}$ the unit vector from receiver to transmitter.

9. Numbers for the midpoint at 100 km altitude, v = 7.5 km/s

Quantity	Value

λ	5.520 m
v/λ	1358.7 Hz
θ (T–P–R bistatic angle)	92.5°
$\lvert\mathbf{b}\rvert = 2\cos(\theta/2)$	1.383
max $\lvert f_D \rvert$	≈ 1879 Hz
k (for ±100 Hz)	0.0532 → ±3.05° of the tangent plane (≈ 5.3 % of directions)